

## DATA CATALOG

### AIONICS HSE DATA WAREHOUSE PROJECT

#### Gold Layer Data Catalog

The Gold layer presents the cleaned Silver data in a star-schema structure designed for reporting, dashboards, and HSE analysis. The dimension views provide descriptive context, while the fact view stores event-level measures and the keys used to relate each event to its dimensions.

| View                     | Type      | Grain                              | Record Key          |
|--------------------------|-----------|------------------------------------|---------------------|
| gold.dim_calendar        | Dimension | One row per calendar date          | calendar_key        |
| gold.dim_cause_reference | Dimension | One row per cause reference record | cause_reference_key |
| gold.dim_equipment       | Dimension | One row per equipment record       | equipment_key       |
| gold.dim_personnel       | Dimension | One row per personnel record       | personnel_key       |
| gold.dim_sites           | Dimension | One row per operating site         | site_key            |
| gold.dim_work_activities | Dimension | One row per work activity          | work_activity_key   |
| gold.fact_event_records  | Fact      | One row per HSE event record       | event_record_key    |

**Design note:** The Gold layer is implemented as views. The calendar dimension uses a dynamically generated ROW\_NUMBER() key, while the other dimension keys are exposed from the cleaned identifiers already available in Silver.

#### 1. gold.dim\_calendar

**Purpose:** Provides calendar attributes for time-based HSE analysis.

**Grain:** One row represents one calendar date.

| Column       | Data Type | Key Type      | Description                                             |
|--------------|-----------|---------------|---------------------------------------------------------|
| calendar_key | BIGINT    | Dimension Key | Unique analytical key generated for each calendar date. |
| date_id      | INT       | Business Key  | Date identifier in YYYYMMDD format.                     |
| full_date    | DATE      | Attribute     | Actual calendar date.                                   |

| Column       | Data Type   | Key Type         | Description                                               |
|--------------|-------------|------------------|-----------------------------------------------------------|
| day_name     | VARCHAR(50) | Attribute        | Name of the day, such as Monday or Tuesday.               |
| day_of_week  | INT         | Attribute        | Numeric weekday value where Monday = 1.                   |
| week_of_year | INT         | Attribute        | ISO week number for the date.                             |
| month_number | INT         | Attribute        | Calendar month number from 1 to 12.                       |
| month_name   | VARCHAR(50) | Attribute        | Calendar month name.                                      |
| quarter      | INT         | Attribute        | Calendar quarter from 1 to 4.                             |
| year         | INT         | Attribute        | Calendar year.                                            |
| is_weekend   | INT         | Attribute / Flag | Indicates whether the date is a weekend: 1 = Yes, 0 = No. |

## 2. gold.dim\_cause\_reference

**Purpose:** Provides standardized immediate and root-cause classifications for HSE events.

**Grain:** One row represents one cause reference record.

| Column                        | Data Type    | Key Type         | Description                                                                |
|-------------------------------|--------------|------------------|----------------------------------------------------------------------------|
| cause_reference_key           | INT          | Dimension Key    | Unique identifier for the cause reference record.                          |
| immediate_cause_category      | VARCHAR(80)  | Attribute        | High-level category of the immediate cause.                                |
| immediate_cause               | VARCHAR(100) | Attribute        | Specific immediate cause associated with an event.                         |
| root_cause_category           | VARCHAR(100) | Attribute        | High-level category of the underlying root cause.                          |
| root_cause                    | VARCHAR(80)  | Attribute        | Specific underlying root cause.                                            |
| human_factor_flag             | INT          | Attribute / Flag | Indicates whether a human factor contributed: 1 = Yes, 0 = No.             |
| equipment_factor_flag         | INT          | Attribute / Flag | Indicates whether equipment contributed: 1 = Yes, 0 = No.                  |
| management_system_factor_flag | INT          | Attribute / Flag | Indicates whether a management-system factor contributed: 1 = Yes, 0 = No. |

### 3. gold.dim\_equipment

**Purpose:** Provides equipment characteristics for analysis of HSE events involving assets.

**Grain:** One row represents one equipment record.

| Column             | Data Type    | Key Type      | Description                                              |
|--------------------|--------------|---------------|----------------------------------------------------------|
| equipment_key      | INT          | Dimension Key | Unique identifier for an equipment record.               |
| asset_id           | NVARCHAR(80) | Attribute     | Operational asset identifier.                            |
| asset_type         | VARCHAR(80)  | Attribute     | Type or category of equipment.                           |
| criticality        | VARCHAR(80)  | Attribute     | Operational criticality classification of the equipment. |
| install_year       | INT          | Attribute     | Year the equipment was installed.                        |
| asset_age_years    | INT          | Attribute     | Equipment age in years.                                  |
| maintenance_status | VARCHAR(80)  | Attribute     | Current maintenance condition or status.                 |
| inspection_status  | VARCHAR(80)  | Attribute     | Current inspection status.                               |
| home_site_id       | INT          | Attribute     | Identifier of the equipment's normal operating site.     |

### 4. gold.dim\_personnel

**Purpose:** Provides personnel characteristics for workforce-related HSE analysis.

**Grain:** One row represents one personnel record.

| Column             | Data Type    | Key Type            | Description                                                 |
|--------------------|--------------|---------------------|-------------------------------------------------------------|
| personnel_key      | INT          | Dimension Key       | Unique identifier for the personnel record.                 |
| employee_id        | NVARCHAR(80) | Attribute           | Employee or contractor identifier.                          |
| employment_type    | VARCHAR(80)  | Attribute           | Identifies whether the person is an employee or contractor. |
| contractor_company | VARCHAR(100) | Attribute           | Contractor company where applicable.                        |
| department         | VARCHAR(80)  | Attribute           | Department associated with the person.                      |
| job_title          | VARCHAR(80)  | Attribute           | Personnel job title.                                        |
| age_group_code     | VARCHAR(80)  | Attribute           | Categorical code representing the person's age group.       |
| years_experience   | FLOAT        | Attribute / Measure | Number of years of work experience.                         |
| training_status    | VARCHAR(80)  | Attribute           | Current training or compliance status.                      |

## 5. gold.dim\_sites

**Purpose:** Provides geographic and operational characteristics of HSE operating sites.

**Grain:** One row represents one operating site.

| Column              | Data Type    | Key Type      | Description                                   |
|---------------------|--------------|---------------|-----------------------------------------------|
| site_key            | INT          | Dimension Key | Unique identifier for the operating site.     |
| site_code           | NVARCHAR(80) | Attribute     | Operational code assigned to the site.        |
| site_name           | VARCHAR(100) | Attribute     | Name of the operating site.                   |
| country             | VARCHAR(80)  | Attribute     | Country where the site is located.            |
| region              | VARCHAR(80)  | Attribute     | Geographic region of the site.                |
| facility_type       | VARCHAR(80)  | Attribute     | Type of facility.                             |
| operation_type      | VARCHAR(80)  | Attribute     | Main type of operation performed at the site. |
| risk_classification | VARCHAR(80)  | Attribute     | HSE risk classification assigned to the site. |

## 6. gold.dim\_work\_activities

**Purpose:** Provides work-activity and risk characteristics associated with HSE events.

**Grain:** One row represents one work activity.

| Column             | Data Type   | Key Type         | Description                                                           |
|--------------------|-------------|------------------|-----------------------------------------------------------------------|
| work_activity_key  | INT         | Dimension Key    | Unique identifier for the work activity.                              |
| activity_category  | VARCHAR(80) | Attribute        | High-level work activity category.                                    |
| activity_name      | VARCHAR(80) | Attribute        | Specific activity being performed.                                    |
| work_type          | VARCHAR(80) | Attribute        | Indicates whether the work is routine or non-routine.                 |
| shift              | VARCHAR(80) | Attribute        | Work shift associated with the activity.                              |
| permit_required    | VARCHAR(80) | Attribute / Flag | Indicates whether a permit is required: Yes or No.                    |
| high_risk_activity | VARCHAR(80) | Attribute / Flag | Indicates whether the activity is classified as high risk: Yes or No. |
| risk_level         | VARCHAR(80) | Attribute        | Risk level associated with the work activity.                         |

## 7. gold.fact\_event\_records

**Purpose:** Central fact view containing HSE event measures and links to the analytical dimensions.

**Grain:** One row represents one HSE event record.

| Column                    | Data Type     | Key Type             | Description                                          |
|---------------------------|---------------|----------------------|------------------------------------------------------|
| event_record_key          | INT           | Fact Key             | Unique identifier for each HSE event record.         |
| event_id                  | NVARCHAR(50)  | Business Identifier  | Human-readable HSE event identifier.                 |
| calendar_key              | BIGINT        | Foreign Key          | Links the event to the calendar dimension.           |
| site_key                  | INT           | Foreign Key          | Links the event to the operating-site dimension.     |
| personnel_key             | INT           | Foreign Key          | Links the event to the personnel dimension.          |
| equipment_key             | INT           | Foreign Key          | Links the event to the equipment dimension.          |
| work_activity_key         | INT           | Foreign Key          | Links the event to the work-activity dimension.      |
| cause_reference_key       | INT           | Foreign Key          | Links the event to the cause-reference dimension.    |
| event_type                | VARCHAR(100)  | Degenerate Attribute | Standardized classification of the HSE event.        |
| severity_class            | VARCHAR(50)   | Degenerate Attribute | Severity classification assigned to the event.       |
| process_safety_tier       | VARCHAR(50)   | Degenerate Attribute | Process-safety tier assigned to the event.           |
| recordable_flag           | INT           | Measure / Flag       | Indicates whether the event is recordable.           |
| lost_time_flag            | INT           | Measure / Flag       | Indicates whether the event is a Lost Time Injury.   |
| fatality_count            | INT           | Measure              | Number of fatalities associated with the event.      |
| days_lost                 | INT           | Measure              | Number of workdays lost.                             |
| restricted_days           | INT           | Measure              | Number of restricted-work days.                      |
| spill_volume_l            | DECIMAL(15,2) | Measure              | Spill or release volume in litres.                   |
| incident_cost_usd         | DECIMAL(15,2) | Measure              | Estimated financial cost of the incident in USD.     |
| downtime_hours            | DECIMAL(15,2) | Measure              | Operational downtime caused by the event, in hours.  |
| regulator_reportable_flag | INT           | Measure / Flag       | Indicates whether the event is regulator-reportable. |

### Star Schema Relationships

The fact view is the centre of the Gold model. Each event record is related to the descriptive dimensions through the keys below.

| Fact Key            | Related Dimension                              | Analytical Meaning                            |
|---------------------|------------------------------------------------|-----------------------------------------------|
| calendar_key        | gold.dim_calendar (calendar_key)               | When the event occurred                       |
| site_key            | gold.dim_sites (site_key)                      | Where the event occurred                      |
| personnel_key       | gold.dim_personnel (personnel_key)             | Who was associated with the event             |
| equipment_key       | gold.dim_equipment (equipment_key)             | Which equipment was associated with the event |
| work_activity_key   | gold.dim_work_activities (work_activity_key)   | What activity was being performed             |
| cause_reference_key | gold.dim_cause_reference (cause_reference_key) | Why the event occurred / contributing causes  |

| Column              | Data Type    | Key Type      | Description                                   |
|---------------------|--------------|---------------|-----------------------------------------------|
| site_key            | INT          | Dimension Key | Unique identifier for the operating site.     |
| site_code           | NVARCHAR(80) | Attribute     | Operational code assigned to the site.        |
| site_name           | VARCHAR(100) | Attribute     | Name of the operating site.                   |
| country             | VARCHAR(80)  | Attribute     | Country where the site is located.            |
| region              | VARCHAR(80)  | Attribute     | Geographic region of the site.                |
| facility_type       | VARCHAR(80)  | Attribute     | Type of facility.                             |
| operation_type      | VARCHAR(80)  | Attribute     | Main type of operation performed at the site. |
| risk_classification | VARCHAR(80)  | Attribute     | HSE risk classification assigned to the site. |

## Calculated fields

### 1. Sales Amount

$$Sales\_amount = quantity \times unit\_price \times (1 - discount)$$

### 2. Allocated freight

$$Allocated\_freight = total\_order\_freight \times \frac{product\_line\_sales}{total\_order\_sales}$$

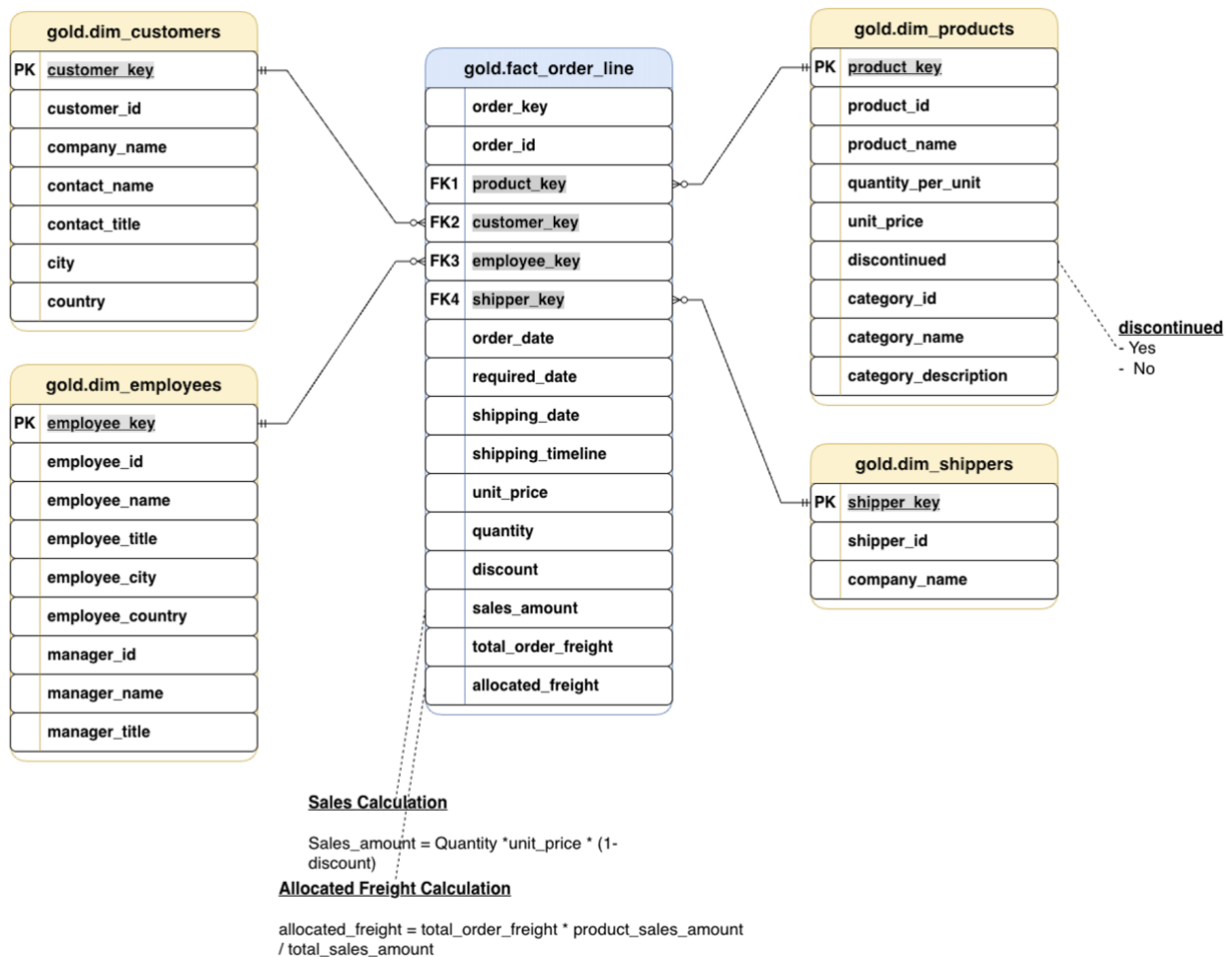
When total order sales (100% discount) are zero, freight is divided equally among the order lines.

### 3. Shipping timeline

| Condition                    | Result           |
|------------------------------|------------------|
| shipped_date < required_date | Shipped Early    |
| shipped_date = required_date | Shipped on Time  |
| shipped_date > required_date | Shipping Delayed |
| shipped_date IS NULL         | N/A              |

## Star-schema relationships

| Fact-table column            | Dimension-table column     | Relationship                     |
|------------------------------|----------------------------|----------------------------------|
| fact_order_line.customer_key | dim_customers.customer_key | Many order lines to one customer |
| fact_order_line.employee_key | dim_employees.employee_key | Many order lines to one employee |
| fact_order_line.shipper_key  | dim_shippers.shipper_key   | Many order lines to one shipper  |
| fact_order_line.product_key  | dim_products.product_key   | Many order lines to one product  |



**Important:** Because the Gold layer uses views, surrogate keys created with ROW\_NUMBER() are dynamically generated and may change when the underlying Silver-layer data changes.